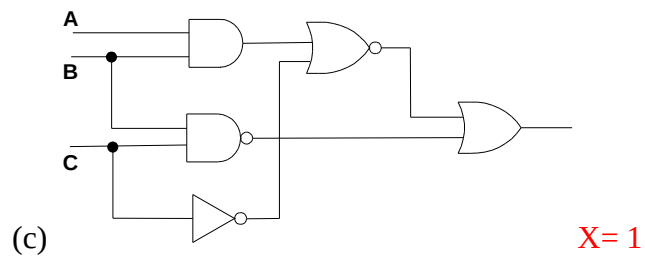
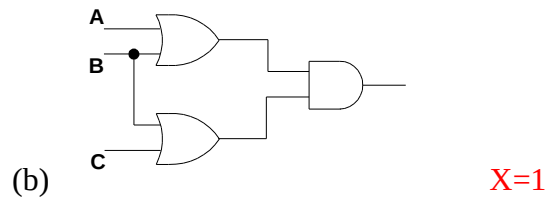
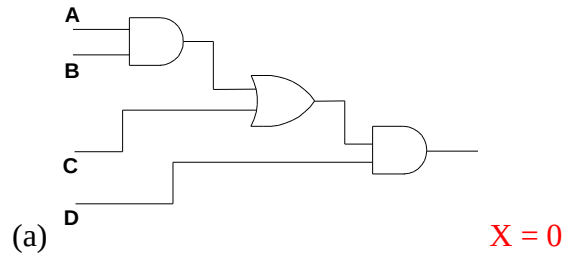


CSC 204 Logic Gates Practice Sheet 1

(Answers are in red)

What would be the output from the following circuits when A=0, B=1, C=1 and D=0



What is the complete truth table for the following? (X' means not X)

(a) $AB + (A+B)$

A	B	AB	A+B	$AB + (A+B)$
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	1

(b) $(BC)' + (AB+C')'$

A	B	C	C'	AB	BC	(BC)'	AB+C'	(AB+C')'	(BC)' + (AB+C')'
0	0	0	1	0	0	1	1	0	1
0	0	1	0	0	0	1	0	1	1
0	1	0	1	0	0	1	1	0	1
0	1	1	0	0	1	0	0	1	1
1	0	0	1	0	0	1	1	0	1
1	0	1	0	0	0	1	0	1	1
1	1	0	1	1	0	1	1	0	1
1	1	1	0	1	1	0	1	0	0

(c) $A' \oplus (AB)$

A	B	A'	AB	$A' \oplus (AB)$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

Which set of logic gates generate the truth table below

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

3 sets of NAND gates